

Minimal building blocks for introspection in self-referential theories

An answer to the minimality question after Corollary 23 of S. A. Alexander, *Self-referential theories* (2020)

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Status. Hand proofs with explicit countermodels. Not machine-checked; no independent review. The literature was not checked for priority.

Source. S. A. Alexander, *Self-referential theories*, Journal of Symbolic Logic 85(4), 2020. [arXiv:2008.11535](https://arxiv.org/abs/2008.11535), [doi:10.1017/jsl.2020.54](https://doi.org/10.1017/jsl.2020.54).

The question. After Corollary 23: “It would be interesting to investigate questions about whether the above building-blocks are minimal. For example, in Lemma 19, is it really necessary to bundle *j*-Introspection with all three other schemas? For now, we will leave those questions open.”

Answer.

- **No: i-Validity is redundant.** $[AV]_i \cup [i\text{-Deduction}]_i \cup [i\text{-Introspection}]_i$ is already closed-r.e.-generic.
- **The other two are both needed.** Dropping Assigned Validity (AV) or *i*-Deduction destroys genericity. So $\{AV, i\text{-Deduction}\}$ in theory *i* is the **unique minimal** bundle that makes *i*-Introspection in theory *j* generic, among subsets of Lemma 19’s three schemas.

Setting (the paper’s §§2-3)

- **The base logic.** The operator T_i is interpreted in a structure by an arbitrary truth value $M \models T_i\phi[s]$, subject only to Definition 2’s three conditions: independence of non-free variables, alphabetic variants, and substitution of *variables*.
- **Validity** means truth in all such structures.
- **The substitution instance** ϕ^s replaces free variables by numerals, including inside operators, so $(T_i\phi)^s \equiv T_i(\phi^s)$.
- **The intended structure** M_U of a family *U* sets $M_U \models T_i\phi[s] \iff U_i \models \phi^s$.
- **Closed-r.e.-generic** means: $M_U \models T$ for every closed (each U_i contains $T_i\sigma$ for $\sigma \in U_i$), r.e. family $U \supseteq T$.
- **The schemas:**
 - AV = $\{\phi^s : \phi \text{ valid}\}$;
 - *i*-Validity (*iV*) = $\text{ucl}(T_i\phi)$ for ϕ valid;
 - *i*-Deduction (*iD*) = $\text{ucl}(T_i(\phi \rightarrow \psi) \rightarrow T_i\phi \rightarrow T_i\psi)$;
 - *i*-Introspection (*iI*) = $\text{ucl}(T_i\phi \rightarrow T_iT_i\phi)$.

The fact that makes AV non-trivial. Validity is *not* closed under substituting numerals inside operators, because Definition 2 links $T_k\phi(x|y)$ with $T_k\phi[s(x|s(y))]$ only for variables *y*.

- *Example.* $\chi(x) \equiv \forall y (y = x \rightarrow (T_k\theta(y) \leftrightarrow T_k\theta(x)))$ is valid, by condition 3.

- But $\chi(n) \equiv \forall y (y = n \rightarrow (T_k\theta(y) \leftrightarrow T_k\theta(n)))$ is not valid: nothing ties the atom $T_k\theta(n)$, with its numeral, to the value of $T_k\theta(y)$ at $y = n$.

What introspection needs

For an assignment s , $M_U \models (T_i\phi \rightarrow T_iT_i\phi)[s] \iff (U_i \models \phi^s \Rightarrow U_i \models T_i(\phi^s))$. Every sentence is some ϕ^s . So:

$M_U \models \mathbf{i\text{-}Introspection} \iff U_i \models \psi \text{ implies } U_i \models T_i\psi, \text{ for every sentence } \psi.$

Theorem 1. $[AV]_i \cup [iD]_i \cup [iI]_j$ is closed-r.e.-generic

It is r.e. by the paper's Theorem 3. Let $U \supseteq$ the family be closed and r.e.

- **The AV and iD parts** are true in M_U , by the paper's Lemmas 18(1) and 17.
- **For iI.** Suppose $U_i \models \psi$. By compactness (Theorem 3(4)), $\chi \equiv \sigma_1 \rightarrow \dots \rightarrow \sigma_n \rightarrow \psi$ is valid for some $\sigma_k \in U_i$.
 - *The key step.* χ is a valid sentence, so $\chi = \chi^s \in AV \subseteq U_i$. Closure then gives $T_i\chi \in U_i$, and also $T_i\sigma_k \in U_i$ for each k .
 - Repeated i-Deduction then gives $U_i \models T_i\psi$. ■

The paper's proof of Lemma 19 obtains $T_i\chi$ from i-Validity. AV already puts the valid *sentence* χ into U_i , and closure supplies $T_i\chi$. So i-Validity is used only to make i-Validity itself true in M_U . That is why Lemma 19 needs it in its own bundle, but introspection does not need it.

Proposition 2. Without i-Deduction, genericity fails

The family is $F = [AV]_i \cup [iV]_i \cup [iI]_j$ (or anything smaller). Take the least closed family $U \supseteq F$, so U_i is the T_i -closure of $AV \cup iV$, plus iI if $i = j$.

The countermodel. Let X be the closure of $U_i \cup \{\chi^s : \chi \text{ valid}\}$ under $\sigma \mapsto T_i\sigma$, taken up to alphabetic variants. Let M have standard first-order part, interpret $T_i\phi[s]$ as " $\phi^s \in X$ ", and interpret every other operator similarly by its own theory.

This is a legitimate base-logic structure. Its truth value depends only on ϕ^s , which gives conditions 1 and 2. And $(\phi(x|y))^s = \phi^s \wedge \{s(x|s(y))\}$, which gives condition 3. Check that $M \models U_i$:

- *Substitution.* M interprets formulas by substitution, so $M \models AV$ (as in Lemma 8 / 18(1)).
- *i-Validity.* $M \models \text{ucl}(T_i\chi)$ for χ valid, since $\chi^s \in X$.
- *Closure members.* $M \models T_i\sigma$ for $\sigma \in U_i$, since $\sigma \in X$, and likewise for iterates.
- *Introspection.* The iI instances hold, since X is closed under T_i .

The failure. Let $\psi \equiv T_i(0=0) \wedge T_i(1=1)$.

- $U_i \models \psi$, since both conjuncts are iV instances.
- But $\psi \notin X$: it is not in U_i , not an instance of a valid formula (its atoms are free), and not of the form $T_i\sigma$. So $M \not\models T_i\psi$, hence $U_i \not\models T_i\psi$.
- By the criterion above, $M_U \not\models \mathbf{i\text{-}Introspection}$. ■

Proposition 3. Without AV, genericity fails

- **If iV is kept** ($[iV]_i \cup [iD]_i \cup [iI]_j$): the iV part is already false in M_U . Take $k \neq i$ and the χ above. Then $\text{ucl}(T_i\chi(x)) \in U_i$, but $M_U \models T_i\chi[x \mapsto n]$ needs $U_i \models \chi(n)$. Here is a model of U_i falsifying $\chi(n)$:
 - make $T_i\phi[s]$ true for every ϕ and s , which satisfies iV , iD , iI and all closure members;

- make $T_k\phi[s]$ true unless ϕ is literally $\theta(n)$.
- This respects Definition 2, whose condition 3 concerns variables only. In it $T_k\theta(y)[y \mapsto n]$ is true while $T_k\theta(n)$ is false, so $\chi(n)$ fails. Hence $U_i \not\models \chi(n)$.
- **If iV is also dropped** ($[iD]_i \cup [iI]_j$, or $[iI]_j$ alone): take the least closed U .
 - $U_i \models 0=0$, trivially.
 - Let D be the closure of U_i under modus ponens and under $\sigma \mapsto T_i\sigma$. U_i contains only iD instances, T_i -prefixed formulas, and iI instances if $i = j$. So every member of D is a T_i -formula or an implication between such, and **$(0=0) \notin D$** .
 - The model $T_i\phi[s] :\Leftrightarrow \phi^s \in D$ satisfies:
 - * the iD instances, because D is closed under modus ponens;
 - * the closure members;
 - * the iI instances, because D is closed under T_i .
 But it makes $T_i(0=0)$ false, so $U_i \not\models T_i(0=0)$.
 - Introspection fails at $\phi = (0=0)$. ■

Corollary

Among the subsets of $\{AV, iV, iD\}$ placed in theory i , those making $[\cdot]_i \cup [iI]_j$ closed-r.e.-generic are exactly those containing **AV and iD**: $\{AV, iD\}$ and $\{AV, iV, iD\}$. The unique minimal one is $\{AV, iD\}$.

Scope, and the second question

- The analysis covers Lemma 19's schemas only. With other schemas in the pool (Peano arithmetic, SMT, the $T_j\phi$ ($\phi \in T_j$) schema) there could be other minimal bundles. **Conjecture G1**: none of Corollary 23's other blocks can replace AV or iD for introspection.
- The stratified analogue (Lemma 64 / Corollary 71) should admit the same reduction: drop the validity schema, keep assigned validity and deduction. It was not checked here.
- The paper's second question, how far the stratified generic construction can be strengthened, is open-ended and untouched.